

LIAM PELIKAN

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EDUCATION

Merrimack College

Master of Science in Computer Science, AI Concentration | GPA: 4.0

North Andover, MA

Dec 2025 - May 2027

University of Florida

Bachelor of Arts in Geography

Gainesville, FL

Aug 2020 - Dec 2024

- **Honors:** UF Presidential Scholarship, National Merit Commended Scholar
- **Coursework:** Machine Learning, Data Structures & Algorithms, Operating Systems, Software Engineering, Databases, Computer Networks, Linear Algebra

SKILLS

- **Languages:** Python, JavaScript, TypeScript, C++, C#, SQL, Java, HTML, CSS, Bash
- **Frameworks:** React, Next.js, Node.js, FastAPI, .NET, WPF, Tailwind CSS, LangChain
- **Data & ML:** Pandas, NumPy, Scikit-learn, Predictive Modeling, Quantitative Analysis, Feature Engineering, RLHF
- **Tools:** Git, Docker, CI/CD, Microsoft Excel (VBA), Data Visualization (Power BI), PostgreSQL, Linux, ArcGIS Pro

WORK EXPERIENCE

Data Annotation

Software Analyst

Remote

Jan 2024 - Present

- Accelerated model evaluation cycles by building internal React and Vite web applications to test AI agent behaviors and constraint adherence.
- Improved LLM code reliability by systematically identifying alignment issues and context-window failures across Python, Java, JavaScript, C++, and SQL training outputs.
- Decreased subjective feedback variance by designing and enforcing comprehensive grading rubrics for complex software engineering coding tasks.
- Reduced logic errors in production deployments by utilizing RLHF and pairwise ranking routines to rigorously debug AI-generated code structures.
- Identified critical algorithmic bottlenecks by designing adversarial test cases targeting multistep logic and framework-based agent reasoning.
- Uncovered underlying quality trends by analyzing platform evaluation metrics and providing actionable, error-reproduction reports to engineering teams.

Northrop Grumman

Software Engineering Intern, Flight Test

Melbourne, FL

May 2021 - Aug 2023

- Cut manual debugging time by 40% by writing automated data validation scripts and Python unit tests for backend switching systems.
- Accelerated data evaluation during flight test campaigns by building WPF/.NET simulation GUIs to quantitatively compare simulated versus real hardware data.
- Enforced CI/CD quality gates by creating a commit tracking tool integrated directly with the Fisheye REST API to monitor repository activity.
- Streamlined test software deployment by maintaining and optimizing continuous integration build environments using Jenkins pipelines.
- Contributed to classified defense program deliverables by collaborating across engineering disciplines on agile workflows across three consecutive internships.

PROJECTS

MovieMatch

Next.js, React, Supabase, PostgreSQL, Clerk Auth, Tailwind CSS

- Full-stack movie recommendation app with swipe discovery (React Spring), real-time watchlist syncing through Supabase, OAuth via Clerk, and social matching to find shared favorites with friends.
- Built a Quantitative Scoring Model using weighted similarity algorithms to cross-reference individual watchlists and rank the most compatible group movie choices based on shared genres and historical ratings.

AI YouTube Shorts Generator

Python, Whisper, LLM API, Docker, FFmpeg, OpenCV

- Automated pipeline that downloads YouTube videos, transcribes them with Whisper, and uses an LLM to pick the best two minute segment for vertical short form content.
- Automatically crops videos and detects faces using OpenCV, burns in subtitles using FFmpeg, runs inside Docker, and outputs a fully rendered video file.

Minesweeper

C++, Raylib, CMake

- Engineered a standalone C++ Minesweeper engine featuring algorithmic guess free board generation, a custom Raylib UI with persistent stat tracking, and a CMake configured Windows release.